

Invariants and Monovariants

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Problems

Problem 1. In the sequence $1, 0, 1, 0, 1, 0, 3, 5, \dots$, each term starting from the seventh is equal to the last digit of the sum of the preceding six terms. Prove that this sequence does not contain six consecutive terms equal to $0, 1, 0, 1, 0, 1$, respectively.

Problem 2. A checker is placed at every lattice point (x, y) with $y \leq 0$. A move consists of jumping one checker over an adjacent checker onto an empty lattice point, then removing the jumped checker. Prove that no checker can ever reach a point (x, y) with $y \geq 5$.

Problem 3. (USAMO 2004) Alice and Bob play a game on a 6×6 grid. On his or her turn, a player chooses a rational number not appearing in the grid and writes it in an empty square of the grid. Alice goes first, and then the players alternate. When all squares have numbers written in them, in each row, the square with the greatest number in that row is colored black. Alice wins if she can then draw a path from the top of the grid to the bottom of the grid that stays in black squares, and Bob wins if she cannot. (If two squares share a vertex, Alice can draw a path from one to the other that stays in those two squares.) Find, with proof, a winning strategy for one of the two players.

Problem 4. (APMO 2007) A regular (5×5) -array of lights is defective, so that toggling the switch for one light causes each adjacent light in the same row and in the same column as well as the light itself to change state, from on to off, or from off to on. Initially, all the lights are switched off. After a certain number of toggles, exactly one light is switched on. Find all the possible positions of this light.

Problem 5. (IMO 1986) To each vertex of a regular pentagon an integer is assigned, so that the sum of all five numbers is positive. If three consecutive vertices are assigned the numbers x, y, z respectively, and $y < 0$, then the following operation is allowed: x, y, z are replaced by $x+y, -y, z+y$ respectively. Such an operation is performed repeatedly as long as at least one of the five numbers is negative. Determine whether this procedure necessarily comes to an end after a finite number of steps.

Problem 6. (IMO 2000) Let $n \geq 2$ be a positive integer and λ a positive real number. We define a move as choosing two fleas at some points A and B , with A to the left of B , and letting the flea from A jump over the flea from B to the point C so that $\frac{BC}{AB} = \lambda$.

Determine all values of λ such that, for any point M on the line and for any initial position of the n fleas, there exists a sequence of moves that will take them all to the position right of M .

Problem 7. (USAMO 2003) A positive integer is written at each vertex of a hexagon. A move is to replace a number by the absolute value of the difference of the two adjacent numbers. If the original numbers sum to 2003^{2003} , prove that it is possible to make a series of moves after which all the numbers are zero.